

BECM 3115

CLIMATE & ARCHITECTURAL DESIGN

SECTION A - MASTER CLASS NOTE

**Exam-oriented | Bangla explanations | Memory techniques | Correct sketches |
Matching final questions**

Prepared from uploaded class notes + Lecture 1-11 slides + Regular Final Question Bank (2016-2024) + Reference Books

Made by 2223033

LECTURE 00

How to Use This Note

এই নোটের উদ্দেশ্য: পরীক্ষায় দ্রুত recall + technically correct answer + sketch-driven presentation

Exam Strategy প্রতি Section-এ ৪ সেট থেকে ৩ সেট উত্তর করতে হবে। তাই Section A-তে “সবকিছু অল্প” না পড়ে ৩টি শক্ত set + ১টি backup set প্রস্তুত করো। এই নোটে Very High / High priority প্রশ্নগুলো সেই উদ্দেশ্যে সাজানো হয়েছে।

Answer-writing pattern (10-20 marks)

1. Definition / core concept - 2 to 3 lines.
2. Key factors / mechanism - bullets or a compact table.
3. Necessary sketch - labelled and relevant, not decorative.
4. Design implication / application - how the concept changes building design.
5. Short conclusion - 1 or 2 lines linking back to comfort / climate responsiveness.

Source discipline

- Primary: Section A Lecture 1-11 slides and uploaded class notes.
- Question matching: Regular Final Term Question Bank 2016-2024 and the student question-analysis sheet.
- Reference check: Manual of Tropical Housing and Building; Climate & Design for the Built-form Designers.

Correction policy যেখানে slide-এর wording বৈজ্ঞানিকভাবে oversimplified/ambiguous, সেখানে “Correction note” দিয়ে exam-safe wording ব্যবহার করা হয়েছে; মূল class framing বজায় রাখা হয়েছে।

LECTURE 01

Environment, Climate, Context & Comfort

মৌলিক ধারণা - যেখান থেকে প্রায় সব design question শুরু হয়

1. Environment and Man-made Environment

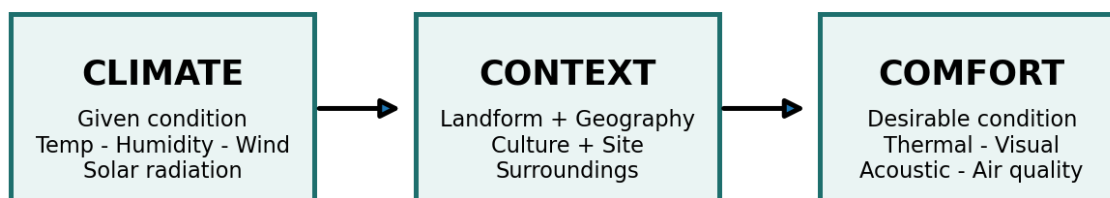
Environment is the overall physical condition in which we live and which can be perceived by the senses. A human abode becomes a man-made environment because people intentionally modify natural conditions to obtain shelter and comfort.

সহজ ভাষায়: প্রকৃতি যেমন আছে সেটি "given"; মানুষ যখন ঘর, জানালা, ছাদ, ছায়া, ventilation ইত্যাদি দিয়ে নিজের জন্য আরামদায়ক অবস্থা বানায়, সেটিই man-made environment.

Term	Exam-ready meaning	সহজ মনে রাখার কৌশল
Environment	Overall condition in which we live; physical surroundings perceived by the senses.	সবকিছু মিলে পরিবেশ
Climate	Long-term environmental/weather condition relevant to heat, humidity, wind, solar radiation etc.	Climate = long-term condition
Man-made environment	Environment modified by humans for shelter, comfort and activity.	Nature + human modification

2. Climate - Context - Comfort

Climate-responsive design = respond to climate + context to achieve comfort



Clean exam sketch: Climate -> Context -> Comfort relationship.

Concept	Main question it answers	Design examples
Climate	What environmental condition is given?	Temperature, humidity, wind, rain, solar radiation
Context	Where and within what physical/cultural	Landform, geography, urban fabric,

	setting are we designing?	culture, local material
Comfort	What desirable human condition must be achieved?	Thermal, visual, acoustic comfort, air quality

Shortcut CCC = Climate is Condition, Context is Circumstance, Comfort is Consequence. বাংলা: অবস্থা - প্রেক্ষাপট - কাঙ্ক্ষিত আরাম।

3. Climatic control process

- Mechanical control - equipment-based heating/cooling/air-conditioning.
- Structural control - orientation, form, material, insulation, shading, envelope.
- Ventilation control - natural or mechanical air movement.

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

- Q.** Write short notes on: (i) Context, (ii) Comfort and (iii) Man-made Environment.
- Q.** Explain “Climate” and “Context” from the point of view of a built-form designer.
- Q.** What differences exist among Climate, Context and Comfort?
- Q.** Can you defend your position about Climate and Built Environment that we should “accept the fate”, “reject the development” or mitigate?
- Q.** What is the difference between Climate and Environment?

LECTURE 02

Climate & Built Environment: Ancient Wisdom, Vitruvius and KHANA

Historical ideas interpreted as climate-responsive design principles

1. Why climate-responsive design is not a new movement

Ancient settlements already used orientation, slope, courtyards, wind protection, solar access and vegetation. The central idea is timeless: a building should respond to the local environment instead of fighting it entirely with mechanical systems.

Aristotle and Xenophon

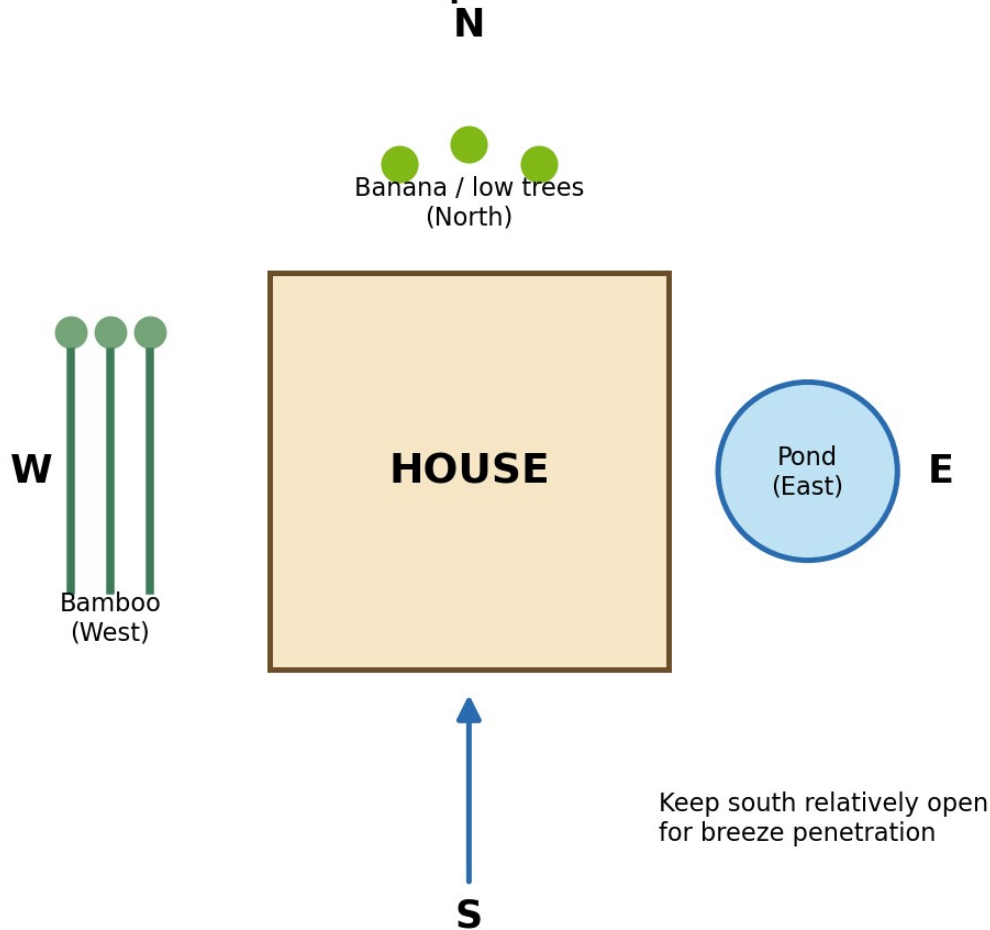
- Aristotle: healthy cities benefit from east-inclined slopes and protection from harsh north winds.
- Xenophon: south-facing arrangements admit winter sun while high summer sun can be shaded by the roof/portico.

Vitruvius - 8 recurring climate-design concerns

No.	Concern	Design meaning
1	Site	Choose healthy site conditions.
2	Internal space	Plan space for appropriate environmental response.
3	Nature vs material	Select material according to natural conditions.
4	Construction vs health	Construction should protect health.
5	Comfort vs temperament	Design for human comfort response.
6	Sustainability	Use resources rationally.
7	City planning	Plan streets/buildings with sun and wind in mind.
8	Connectivity	Movement and environmental connection affect comfort.

2. KHANA maxim - vernacular climatic interpretation

Vernacular climatic interpretation of KHANA maxim



Khana-based site zoning - use as a simplified exam sketch, not as a rigid modern code.

Element	Traditional placement	Climatic interpretation
Pond / "duck"	East	Water + evaporation; earth from pond can raise plinth; traditional environmental treatment.
Bamboo	West	Dense vegetation helps shade low-angle west sun.
Banana / low vegetation	North	Low-height planting does not strongly obstruct south-north air movement; north gives diffuse light.
Open / low obstruction	South	Allows southern breeze and ventilation penetration.

Exam caution Khana's maxim is a vernacular planning interpretation, not a universal scientific rule. লিখবে: "in the climatic and cultural context of Bengal/Bangladesh".

3. “Accept fate / reject development / mitigate”

The climate-design position is mitigation: development should continue, but form, orientation, materials, openings, landscape and technology should reduce environmental stress and energy dependence.

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

- Q. According to KHANA, “The south facing house is by far the best...” - do you agree? Justify with examples and sketches.
- Q. Show the zoning of a building in a particular site according to KHANA for a climate-responsive building in Bangladesh.
- Q. Write down the suggestions of Vitruvius on climate and architecture.
- Q. Can you defend the position that Climate and Built Environment should mitigate rather than simply accept fate or reject development?

LECTURE 03

Climatic Comfort & Human Heat Exchange

Section A-এর সবচেয়ে repeated theory cluster

1. Factors of climatic comfort

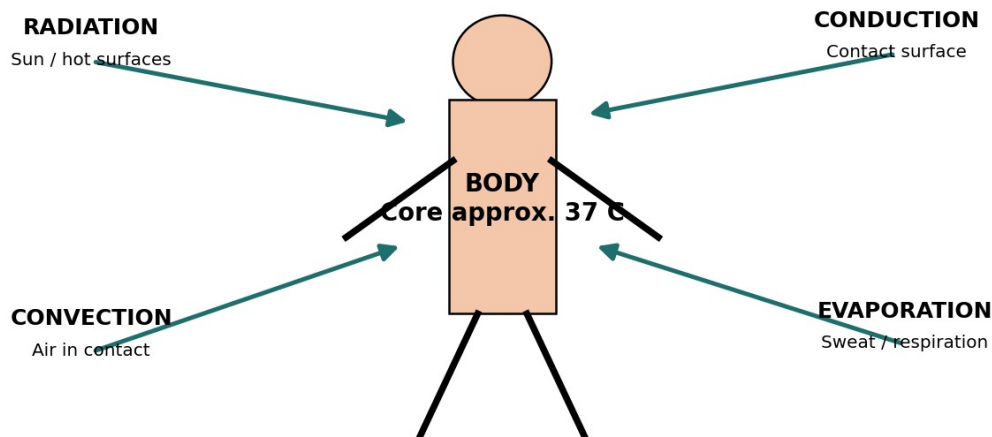
Factor	What it changes	Bangla cue
Air temperature	Sensible heat gain/loss by convection.	বাতাস কত গরম/ঠান্ডা
Radiation / surrounding surface temperature	Radiant exchange with sun, walls, roof, hot/cold objects.	চারপাশের surface heat
Humidity	Controls evaporation potential.	ঘাম শুকাবে কি না
Air movement	Changes convection and evaporation.	বাতাস চলাচল

Memory T-R-H-A = Temperature - Radiation - Humidity - Airflow. মনে রাখো: “তাপ-রশ্মি-আর্দ্রতা-বাতাস”.

2. Human thermal comfort

The human body continuously produces heat through metabolism. Thermal comfort requires a steady thermal state in which surplus heat is dissipated without excessive physiological strain.

Human heat exchange: heat can be gained or lost depending on temperature gradients



Core mechanisms: conduction, convection, radiation and evaporation.

Heat-balance equation - exam-safe form

$$M - W = C + R + K + E + S$$

M = metabolic heat; W = useful mechanical work; C = convection; R = radiation; K = conduction; E = evaporation; S = body heat storage. At steady state, S approx. 0.

সহজ ভাষায়: শরীর যত net heat তৈরি করে, প্রায় ততটাই পরিবেশে ছাড়তে পারলে body temperature steady থাকে। জমে গেলে S positive -> শরীর গরম হয়; বেশি হারালে S negative -> শরীর ঠান্ডা হয়।

3. Deep body / skin temperature and physiological response

Core body temperature is maintained close to 37 C. The lecture material treats skin temperature as roughly 31-34 C in hot-condition heat-exchange discussion. Vasomotor adjustment changes skin blood flow to regulate heat transfer.

Condition	Key idea	Design relevance
Cold	Vasoconstriction reduces skin blood flow; shivering may increase metabolic heat.	Reduce drafts; conserve heat.
Warm/hot	Vasodilation increases skin blood flow; sweating supports evaporative cooling.	Provide air movement and evaporation opportunity.
Hot + saturated + still air	Evaporation becomes ineffective; body heat may accumulate.	Dangerous: maximize ventilation, reduce radiant load.

4. Heat loss in four climatic situations

Situation	Dominant process	Exam point
Calm warm air + moderate RH	Radiation + convection + evaporation	Lecture example: approx. 45% radiation, 30% convection, 25% evaporation.
Hot air + considerable radiation	Convection/radiation loss decreases as air/surface temperatures approach skin temperature.	Evaporation becomes increasingly important if air is dry enough.
Hot air + radiation + air movement	Moving air removes saturated film around skin and accelerates evaporation.	Air movement is a key relief mechanism.
Air warmer than skin + saturated + still	Very little sensible or evaporative heat loss.	Heat-stroke risk; potentially fatal conditions.

Correction note Higher air speed increases evaporation and usually improves evaporative cooling when the air is not saturated. It does not automatically increase ambient relative humidity; the key exam idea is “moving air removes the saturated boundary layer at the skin”.

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

- Q. What are the factors of Climate Comfort? Explain with examples.
- Q. Describe the processes of heat loss in various climatic situations.
- Q. Explain the human body heat exchange process with necessary sketches.
- Q. Describe human body heat exchange with its environment through conduction, convection, radiation and evaporation.
- Q. What are the thermal comfort ideas?
- Q. Specify the heat exchange process for a built environment with necessary sketch and example.
- Q. What is the thermal balance equation?

LECTURE 04

Designer Responsibilities, Essential Knowledge & Complexities

How a built-form designer translates climate data into design decisions

1. Designer responsibilities

External responsibilities	Internal responsibilities
Minimum interference to useful normal wind flow.	Ensure adequate oxygen-rich air.
Control adverse solar impact / avoid harmful deep shade where not desirable.	Maintain conducive indoor temperature.
Manage precipitation and rain exposure.	Maintain conducive humidity and air quality.

2. Essential knowledge

- Science of climate - temperature, radiation, wind, humidity and rainfall.
- Physiology - how the human body responds to heat/cold.
- Human comfort - acceptable ranges and subjective variables.
- Building materials - thermal capacity, insulation, moisture response, durability.
- Passive design technology - orientation, shading, ventilation, form, landscape.
- Active design technology - mechanical heating/cooling/ventilation when passive measures are insufficient.

3. Design complexities

Complexity	Why difficult
Extreme conditions	Comfort may not be achievable by openings alone.
Metabolic heat	Occupancy/activity adds internal heat.
Energy + cost	Comfort must be balanced with operating cost and maintenance.
Composite climate	Design may need opposite strategies in different seasons.
Material availability	Local cost, durability and construction skill affect feasibility.

Shortcut RESPONSIBILITY = outside climate filter + inside comfort keeper. ESSENTIAL KNOWLEDGE = climate + body + material + passive + active.

MATCHING QUESTION BANK [HIGH PRIORITY]

- Q. Explain the essential knowledge for Climate Design.
- Q. What are the responsibilities of the built-form designers? What kind of complexities do they have to face in Climate Design?
- Q. Discuss the designers' responsibilities in ensuring thermal comfort in built environment.

LECTURE 05

Comfort Indices, Cosine Law, Atmospheric Depletion, Humidity & Air Movement

Definitions, differences and short-note answers

1. Subject variables affecting comfort

Comfort is not determined by climate alone. Clothing, acclimatization, age and sex, body shape/fat, health, food/drink and other personal factors modify thermal sensation.

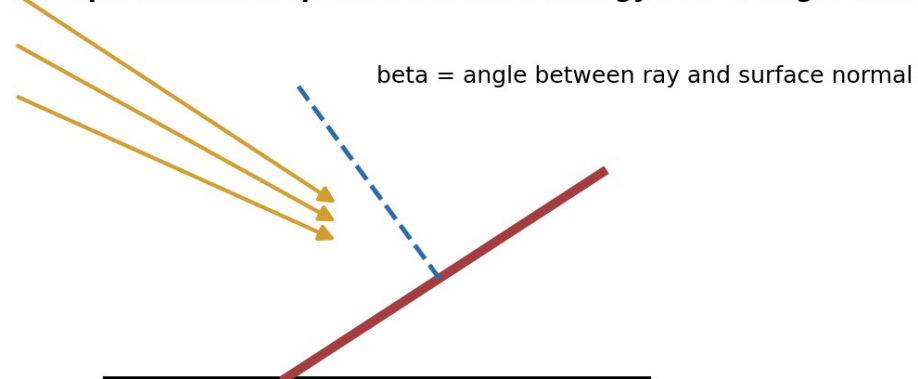
2. ET, CET, EW and Operative Temperature

Index	Combines	Core exam definition / limitation
ET - Effective Temperature	Air temperature + humidity + air movement	Equivalent sensation scale; originally based on equal-comfort lines. Radiation is not directly included.
CET - Corrected Effective Temperature	ET variables + radiation effect	ET corrected to account for radiant environment; widely used in the course material.
EW - Equivalent Warmth	Air temp + humidity + mean radiant temp + skin/clothing surface response	Can underestimate cooling effect of air movement at high humidity.
OT - Operative Temperature	Air temperature + mean radiant temperature	Useful where humidity and air movement effects are small/negligible.

Memory ET = 3 variables; CET = ET + radiation. OT = Air + Radiation. EW = “whole warmth response” including body/clothing surface measurements.

3. Cosine Law

Cosine Law - oblique incidence spreads the same energy over a larger area



$$I_{\text{tilted}} = I_{\text{normal}} \times \cos(\text{beta})$$

Angle of incidence is measured from the surface normal.

$$I_{\text{tilted}} = I_{\text{normal}} \times \cos(\beta)$$

β = angle between the incident ray and the normal to the surface. More oblique incidence $\rightarrow$ lower intensity per unit area.

4. Atmospheric depletion

Solar radiation is reduced while passing through the atmosphere by absorption and scattering due to ozone, water vapour, dust, clouds and other particles. A longer atmospheric path generally means greater depletion.

সহজ ভাষায়: সূর্যের রশ্মি atmosphere-এর ভিতর যত বেশি পথ যায়, তত বেশি অংশ absorb/scatter হয়। তাই low sun angle-এর radiation বেশি depleted হতে পারে।

5. Humidity terms

Term	Meaning	Exam note
Saturated air	Contains the maximum water vapour possible at that temperature.	RH = 100% at saturation.
Unsaturated air	Contains less vapour than saturation capacity.	Evaporation is possible.
Supersaturated air	Contains more vapour than equilibrium saturation capacity.	Unstable; condensation tends to occur.
Absolute Humidity (AH)	Actual water vapour amount per unit mass/volume (as defined in course material).	Used in RH calculation with matching units.
Relative Humidity (RH)	AH / saturation humidity at same temperature x 100%.	Dimensionless percentage.

$$RH = (AH / SH) \times 100\%$$

AH and SH must be expressed on a consistent basis at the relevant temperature.

6. Factors affecting humidity and air movement

Humidity level factors	Air movement factors
Soil condition	Unequal solar heating
Vegetation	Earth rotation / Coriolis effect
Altitude	Pressure differences
Water body	Global / regional wind belts
Air movement	Local topography and obstruction
Precipitation	Opening and building layout at micro scale

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

Q. Explain with necessary sketches and examples: (i) ET, (ii) CET and (iii) Bioclimatic chart.

- Q. Write short notes on: Cosine Law, Atmospheric Depletion, Stevenson Screen, Super-saturated Air and Corrected Effective Temperature.
- Q. Discuss temperature, humidity and air movement as factors of climate comfort.
- Q. What are effective temperature and corrected effective temperature? Show the differences.
- Q. Differentiate among effective temperature, corrected effective temperature and equivalent warmth.
- Q. Explain atmospheric depletion and Cosine Law with necessary sketches.

LECTURE 06

Regional Climate, Site Climate & Measuring Climatic Data

Factors + instruments + site-design logic

1. Global, regional and site climate

Scale	Meaning	Example focus
Global	Large climatic belt / global circulation context.	Tropical, temperate etc.
Regional	Large geographic area below global scale but above site scale.	Location, altitude, diurnal variation, terrain.
Site / microclimate	Local climate of the actual design site.	Soil, windbreaks, water body, urban surfaces, immediate topography.

2. Factors constituting regional climate

Group	Factors
Primary	Global climatic general factors.
Secondary	Location of region, diurnal variation, altitude.
Tertiary	Soil, topography, natural/man-made windbreaks, water bodies, cities/industries, deserts.
Local & minor	Ocean current, sky condition, smog, smoke, fog, thunder, sand storm etc.

Memory P-S-T-L = Primary - Secondary - Tertiary - Local. বাংলা cue: “প্রধান-সহায়ক-তৃতীয়-লোকাল”.

3. Factors constituting site climate

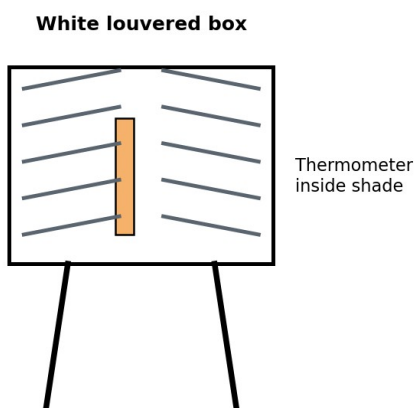
- Soil condition in the locality.
- Topography of the site.
- Small hill/rock acting as windbreak.
- Man-made windbreak.
- Small water bodies.
- Cities and industries / surrounding built-up condition.

4. Measuring climate

Parameter	Instrument / method	Unit / exam note
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Air temperature	Dry-bulb thermometer / electronic sensor; ideally in Stevenson Screen	degree C
Humidity	Wet and dry-bulb hygrometer / electronic hygrometer; psychrometric relation	% RH, AH in g/kg or g/m ³
Air movement	Anemometer	m/s
Long-term data	Daily/monthly max-min-mean and extremes	Designer needs both average and extreme values

Stevenson Screen - measures true air temperature away from direct sun/rain



Typical mounting height: about 1.2-1.8 m above ground

Stevenson Screen - exam-ready schematic.

Stevenson Screen-এর মূল কাজ: thermometer-কে direct sun, rain এবং local radiation থেকে protect করে shaded, ventilated অবস্থায় true air temperature মাপা।

5. Large-site design process

6. Start from regional climatic data.
7. Identify dominant local modifiers - topography, water, vegetation, built mass, windbreaks.
8. Estimate site deviations from the regional condition.
9. Select the most suitable habitation zone.
10. Translate climate findings into orientation, layout, materials, openings and landscape.

MATCHING QUESTION BANK [HIGH PRIORITY]

- Q. What are the tertiary factors constituting the regional climate?
- Q. Show the working mechanism of Stevenson Screen for measuring temperature.
- Q. In which methods can you measure temperature, humidity and air movement for designing a building in a particular climate?
- Q. Show the factors constituting regional climate and site climate.
- Q. Describe the fundamental process of designing a site as a designer using site-climate knowledge in the case of a large site.
- Q. Explain the five factors that contribute to the formation of a site climate.

LECTURE 07

Design Specifications: Layout, Spacing, Air Movement & Enclosure

The design-answer chapter: always draw

1. Design specification checklist

The lecture organizes climate-responsive design around five practical controls: Layout, Spacing, Air movement, Enclosure and Shading Device.

Shortcut L-S-A-E-S = Layout - Spacing - Air movement - Enclosure - Shading. মনে রাখো: “লেআউট-স্পেস-বাতাস-খোলস-ছায়া”.

2. Layout for wind and climate

- Broad spacing can improve breeze penetration in warm-humid climates.
- Staggered layout often avoids wind shadows and improves airflow through a group of buildings.
- Building orientation can be adjusted about 30-45 degrees to wind direction where suitable to improve exposure and flow distribution.
- In hot-dry/composite cases, compactness may reduce exposed surface area and unwanted heat gain; design must respond to season and climate type.

3. Spacing and functional distribution

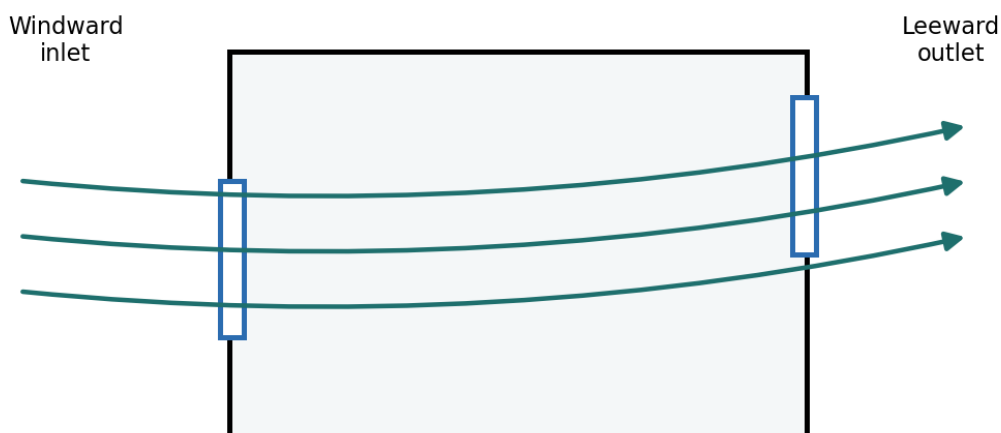
- Organize functions according to heat, privacy, day/night use and air movement.
- Use floor height and internal voids/courtyards to support air movement and thermal stratification.
- Separate served and service zones where this helps buffer unwanted heat.

P/A ratio

For a given floor area, a compact form generally has a lower perimeter-to-area (P/A) ratio, which reduces exposed envelope area. Circular/compact geometries have low P/A ratios, while highly articulated forms have higher exposure.

4. Air movement and openings

Cross ventilation: openings on opposite / adjacent pressure zones



Good practice: keep airflow path unobstructed; outlet can be similar or slightly larger than inlet

Cross ventilation - a high-value exam sketch.

Opening decision	Effect
Position	Determines pressure difference and airflow path through occupied zone.
Size	Controls flow area; too-small openings restrict ventilation.
Control / louvers	Can direct air upward/downward and regulate rain/privacy.
Cross ventilation	Requires connected inlet-outlet path across the space.

5. Enclosure

- Material and thermal capacity determine heat storage and time lag.
- Degree of enclosure balances protection against the need for airflow.
- Insulation, roof build-up and external surface reflectance can reduce heat entry.

6. Warm-humid vs hot-dry design

Warm-humid	Hot-dry
Maximum ventilation with large openings.	Control hot dusty wind; selective ventilation.
Maximum shading of direct and diffuse radiation.	Compactness / shaded external spaces often beneficial.
Avoid heat storage where possible.	Use thermal mass and time lag strategically.
Reflective surfaces, ventilated double roof.	Courtyard / evaporative cooling can help where water is available.
Vegetation should shade but not block useful breeze.	Protect against glare and long-wave re-radiation from surrounding hardscape.

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

- Q.** Discuss layout, spacing and position of opening for ensuring air movement and ventilation in a building. Provide necessary sketches.
- Q.** Provide appropriate design solution with necessary sketches for a naturally ventilated building in composite monsoon climate.
- Q.** Discuss the microclimatic solution for hot-dry and warm-humid climate in terms of spacing. Provide necessary sketches.

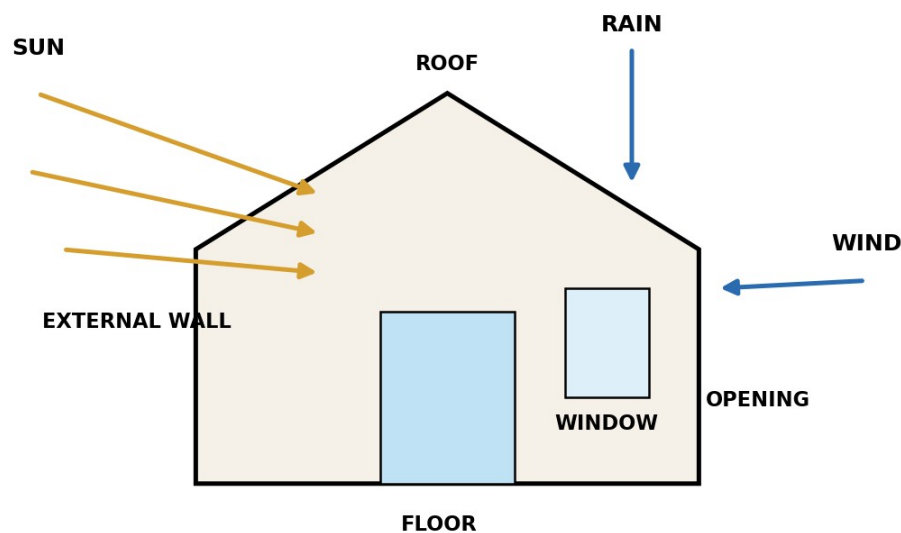
LECTURE 08

Designing Microclimate: Building Envelope & Design Tools

Roof, floor, wall, window, door, balcony, terrace, overhang, clerestory

1. What is being moderated?

Microclimate design treats each part of the building envelope as an environmental filter. Heat, humidity, solar radiation, rain and air movement affect components differently, so the designer selects component-specific responses.



Microclimate design moderates heat, moisture, solar radiation and airflow at the envelope

Envelope-focused microclimate sketch.

2. Component-wise design solutions

Component	Main climate problem	Design response
Roof	Solar heat gain; time lag; moisture penetration	Reflective finish, insulation, air cavity/double roof, ventilation near roof, waterproofing/membrane.
Ground floor	Heat exchange with ground; damp	Insulation where needed; DPC / moisture-resistant membrane.
Intermediate floor	Heat/moisture transfer between floors	Depends on material conductance and temperature/humidity difference.
External wall	Solar heat, rain, orientation-dependent exposure	Material/thickness/color/texture, insulation, overhang, rain-shedding detail.
Internal wall	Usually lower temperature difference	Often functions as internal thermal separator/insulator.
Window	Direct solar gain, infiltration, rain/fog	External shading, suitable glazing, tightness

		where needed, controlled opening.
Ventilator	Exhaust of warm air + air change	Place to assist high-level exhaust / pressure-driven flow.
Door	Infiltration and access	Weather-sealing if needed; use strategically for airflow.
Balcony / overhang	Sun + rain on wall/window	Projection shades wall/opening and reduces rain exposure.
Terrace	Solar heating and re-radiation	Cool/reflective finish, insulation, drainage, shading/planting where appropriate.
Clerestory	High-level light + ventilation	Useful for daylight and stack-assisted exhaust.

3. Design tools for moderating microclimate

Tool	What the designer manipulates
Materials	Type, thickness, composition, consistency, colour, texture.
Orientation	Built-form orientation in relation to sun/wind.
Openings	Size, position, fragmentation, controls and shading.
Architectural design	Form, section, roof, envelope, courtyards.
Micro-planning	Assigning functions to suitable zones.
Semi-mechanized techniques	Fans/controlled aids when passive design alone is insufficient.

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

- Q.** Can you explain how climate factors affect the building envelopes in designing microclimate? Use necessary sketches.
- Q.** Mention the design tools for moderating microclimate.
- Q.** In designing microclimate, how are different components of a building affected by climatic factors? Show the design solution.

LECTURE 09

Shading Devices & Effective Shading

Solar control with orientation-sensitive sketches

1. Why shading?

The principal purpose of sun shading is to prevent unwanted solar radiation when external conditions are above comfort level. Shading can also help control glare, rain exposure, privacy/visibility and sometimes wind direction.

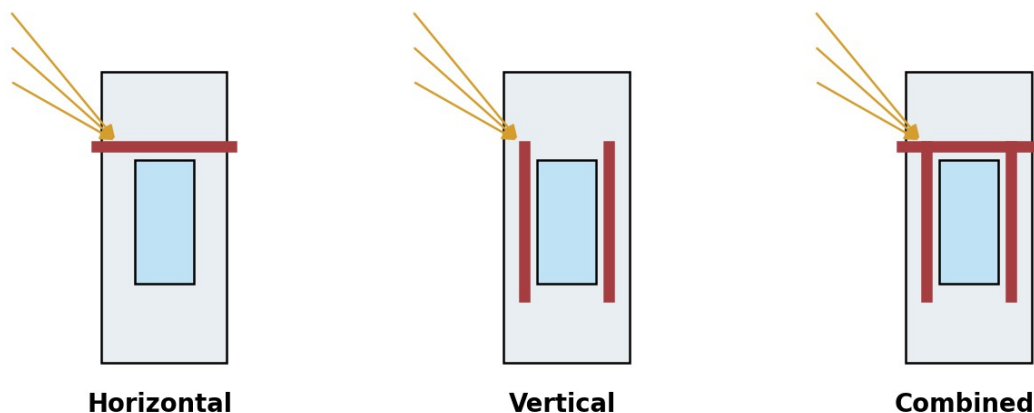
Solar path into building	Meaning
Direct radiation	Beam radiation directly through an opening.
Reflected radiation	Solar radiation reflected from ground/adjacent surfaces.
Diffuse radiation	Scattered sky radiation, important in cloudy/warm-humid conditions.
Indirect through opaque surface	Solar heat absorbed by roof/wall and later conducted/radiated inward.

2. External vs internal shading

External shading is generally more effective for heat control because it stops solar radiation before it passes through the glazing and becomes indoor heat. Internal curtains can control glare but the absorbed solar heat is already inside the envelope.

3. Types

Basic shading-device types



Basic types: horizontal, vertical and combined shading.

Type	Best suited conceptually for	Examples
Horizontal	High-angle sun; often useful on south-facing exposures in many northern-hemisphere situations.	Overhang, canopy, horizontal louvers.

Vertical	Low-angle sun from east/west directions.	Vertical fins, side screens.
Combined	Where both vertical and horizontal control are needed.	Egg-crate, combined fins + overhang.

Exam sketch tip Sketch-এ অবশ্যই sun direction, window, shading projection এবং blocked ray দেখাও। শুধু fin-এর shape আঁকলেই full mark পাওয়ার সম্ভাবনা কম।

MATCHING QUESTION BANK [HIGH PRIORITY]

- Q. What is effective shading? Explain with sketches.
- Q. Specify various types of shading devices with necessary sketches.
- Q. How many ways can you ensure climate comfort in indoor environment of a building by designing shading devices?

LECTURE 10

Climate-friendly Movements: Tube House, Organic Design, Rat-trap Bond & Green Building

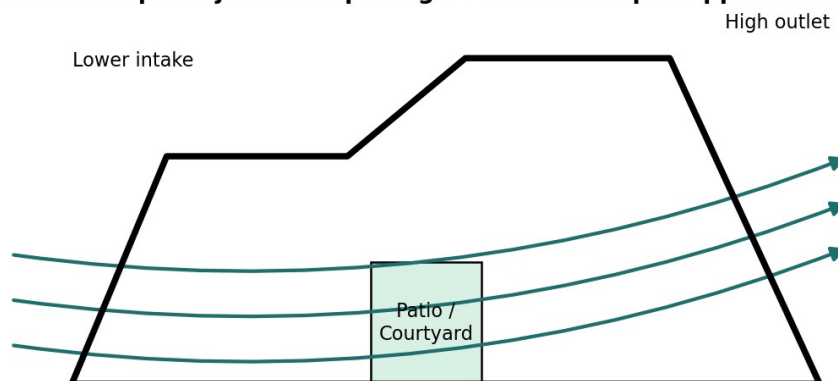
Short notes + construction detail + sustainability

1. Climate-friendly movement topics

Topic	Core idea
Tube house	Section/form and adjustable openings use natural air flow and changing day-night requirements.
Organic architecture	Form responds to nature, airflow, site and material; cross ventilation and landscape integration.
Climate-responsive design	Building form and envelope respond directly to local climate.
Sustainable building	Balances environmental, social and economic performance over life cycle.
Green building	Reduces measurable impacts such as energy, water, waste and indoor-health problems.
Contemporary materials/technology	Use technology without losing climate response, durability and life-cycle logic.

2. Tube house

Tube-house concept: adjustable openings + section shape support natural ventilation



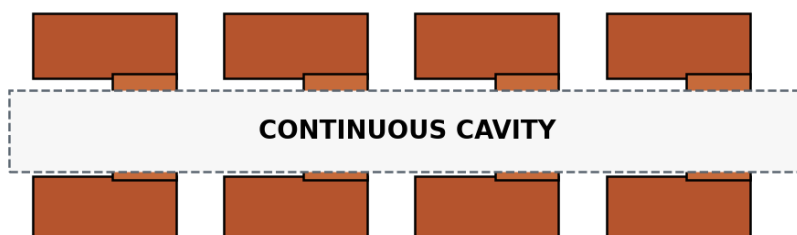
Simplified tube-house ventilation idea for exam use.

সহজ ভাষায়: Tube House-এর section এমনভাবে কাজ করে যাতে নিচু দিক দিয়ে fresh air ঢোকে, inner court/patio দিয়ে circulate করে এবং উঁচু opening দিয়ে hot air বের হতে পারে; day/night use-ও spatially adjust করা যায়।

3. Rat-trap bond

Rat-trap bond places bricks on edge so that a cavity is created within the wall thickness. The lecture emphasizes reduced material use, lower cost and improved thermal/sound insulation when correctly detailed.

Rat-trap bond - bricks on edge create a cavity while retaining wall thickness



Exam sketch: show brick leaves, cavity, solid courses at sill/lintel/top/bottom

Rat-trap bond conceptual plan / cavity sketch.

Construction rule	Why it matters
Use consistent, good-quality bricks	Bond geometry depends on dimensional consistency.
Bottom and top courses solid	Stability and closure of cavity.
Sill/lintel and opening jamb zones solid	Allows frame fixing and local strength.
Reinforcement at corners/openings where designed	Improves integrity / earthquake detailing.
Plan services through cavity carefully	Improves finish but must not weaken wall.

Correction note The slide states wall strength is “not compromised”. Exam-safe wording: properly designed and constructed rat-trap masonry can provide adequate wall performance, but structural capacity depends on brick quality, workmanship, detailing and applicable code requirements.

4. Green vs Sustainable building

Green building	Sustainable building
Focuses on measurable reductions in water, energy, waste and indoor environmental impacts.	Broader life-cycle goal balancing environment, society and economy.
Can be “greener than baseline” without solving every life-cycle issue.	Requires long-term viability and wider system thinking.
Typical actions: efficient energy/water, resource conservation, healthier indoor spaces.	Includes resilience, affordability, social equity, durability and environmental performance.

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

- Q.** Explain with examples: (i) Tube House, (ii) Organic Architecture, (iii) Green building movement and (iv) Contemporary building materials.
- Q.** Define Green Building Movement.
- Q.** Evaluate and compare contemporary and traditional building materials from the climate point of view.
- Q.** Distinguish between sustainable building and green building.
- Q.** Show the construction techniques and advantages of a brick wall using Rat-Trap bond. Which walls can you design with rat-trap bond to ensure thermal comfort?
- Q.** Show the working mechanism of a tube-house air ventilation.
- Q.** Illustrate the working mechanism of rat-trap bond in brick bonding.

LECTURE 11

Non-thermal Microclimate: Ventilation, Air Space, Infiltration & Numericals

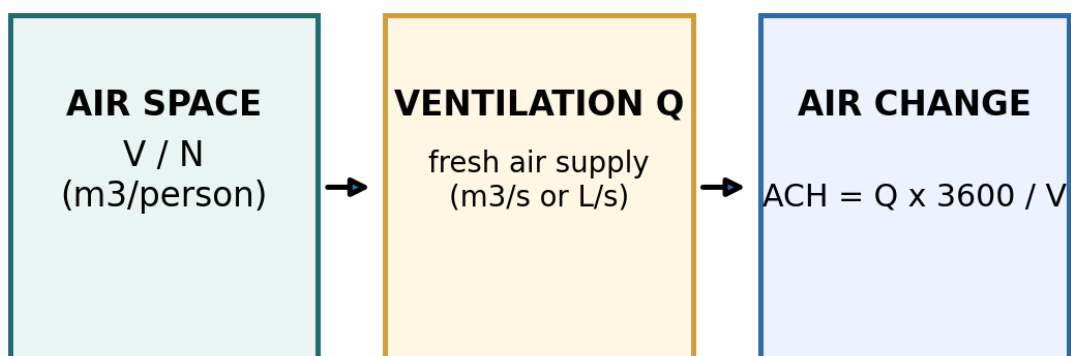
Formula-based preparation - do not leave this chapter for the exam hall

1. Objectives of ventilation / air change

- Supply adequate oxygen for breathing and other purposes.
- Remove carbon dioxide.
- Remove unwanted heat and moisture.
- Reduce staleness and smell / pollutants.

2. How ventilation rate is expressed

Ventilation calculation chain



Core numerical relationship.

$$\text{ACH} = (Q \times 3600) / V$$

Q in m3/s; V = room volume in m3. If Q is in L/s, first divide Q by 1000.

Air space per person = (net room volume) / number of users

Net room volume may subtract major furniture/occupant volume if the problem specifically requires it.

3. Recommended air space and fresh-air supply (course table)

Air space/person (m3)	Minimum (L/s/person)	Smoking zone	Non-smoking zone
3	11.3	22.6	17.0
6	7.1	14.2	10.7

9	5.2	10.4	7.8
12	4.0	8.0	6.0

Table-reading rule প্রথমে room volume / person থেকে air-space category নির্ধারণ করবে; তারপর smoking/non-smoking condition দেখে L/s/person নেবে; শেষে total Q = rate x number of persons.

4. Relative humidity numerical

$$RH = (AH / SH) \times 100\%$$

Example: AH = 12 g/m³, SH = 17 g/m³ -> RH = 70.6%.

5. Ventilation numerical - worked example

Example: room 45 m x 23 m x 3 m, 200 persons, non-smoking. Volume V = 3105 m³, so air space/person = 15.525 m³. Using the highest listed 12 m³ category, the course table gives 6.0 L/s/person for a non-smoking zone.

11. Required fresh air = 200 x 6.0 = 1200 L/s = 1.20 m³/s.
12. ACH = 1.20 x 3600 / 3105 = 1.39 air changes/hour (approx.).

Exam caution If the teacher expects interpolation or a different table-selection convention, follow the convention demonstrated in class. The uploaded table itself ends at 12 m³/person.

6. Heat loss due to air change

$$\text{Heat loss rate} = \rho \times Q \times c_p \times (T_{\text{inside}} - T_{\text{outside}})$$

Standard engineering form. If ACH is given: $Q = \text{ACH} \times V / 3600$. ρ is air density, c_p is specific heat.

Standard engineering note যদি question-এ air density না দেওয়া থাকে, engineering calculations often use about 1.2 kg/m³ near room conditions. এটি slide table-এর অংশ নয়; exam-এ teacher/class convention অনুসরণ করবে।

7. Fire shelter / laboratory safety logic

13. Calculate total room volume.
14. Calculate air space per person.
15. From the course table, find required fresh air per person for the occupancy condition.
16. Compare required total fresh air with available supply.
17. If inadequate, improve by increasing fresh-air supply, increasing volume, reducing occupancy, or improving the ventilation arrangement.

MATCHING QUESTION BANK [VERY HIGH PRIORITY]

Q. Define Absolute Humidity (AH) and Saturation Point Humidity (SH) and calculate Relative Humidity (RH) from given vapour densities.

- Q.** What does it mean by air change? Mention how the rate of ventilation is expressed in a built-space.
- Q.** A room containing many people is given. Find required ventilation in m^3/s , L/s and air change per hour.
- Q.** What does it mean by air space and infiltration rate? What are the heat losses due to air change in a building?
- Q.** A fire shelter is designed for a given number of persons. Predict the safety situation and suggest how it can be made safer.
- Q.** A laboratory has a given fresh-air infiltration rate. Find whether it is suitable for the stated number of users.

LECTURE R

Rapid Revision: Formulas, Sketches & Last-minute Checklist

শেষ ৩০-৬০ মিনিটে শুধু এই অংশ দেখলেও core recall হবে

1. Formula sheet

Formula	Use
$RH = (AH / SH) \times 100\%$	Relative humidity numerical.
$ACH = Q \times 3600 / V$	Convert ventilation flow rate to air changes/hour.
$Q = ACH \times V / 3600$	Convert air changes/hour to m ³ /s.
Heat loss = $\rho \times Q \times c_p \times \Delta T$	Ventilation heat-loss rate.
$I_{\text{tilted}} = I_{\text{normal}} \times \cos(\beta)$	Cosine-law solar intensity on tilted surface.
$M - W = C + R + K + E + S$	Human heat balance; steady state S approx. 0.

2. Must-practice sketches

Sketch	Why it matters
Climate - Context - Comfort flow	Very easy marks in definition/comparison questions.
KHANA zoning	Frequently asked with Bangladesh context.
Human heat exchange	Core thermal-comfort answer.
Cosine-law incidence	Short-note / explanation.
Stevenson Screen	Measurement question.
Cross ventilation	Layout/opening/design specification.
Envelope climate effects	Microclimate question.
Horizontal / vertical / combined shading	High-frequency shading question.
Tube-house ventilation	Climate-friendly movement short note.
Rat-trap bond cavity	Construction + thermal comfort.

3. 3-set preparation recommendation

Set A - safest theory cluster Lectures 1-5: CCC, KHANA/Vitruvius, thermal comfort, heat exchange, ET/CET, designer responsibility.

Set B - design + sketch cluster Lectures 7-9: layout, spacing, openings, microclimate envelope, shading.

Set C - construction + numerical cluster Lectures 10-11: rat-trap / green / tube house + ventilation / air-space / RH numericals.

Backup Lecture 6 regional/site climate + measurement is a strong backup because answers are structured and sketch/table friendly.

4. Common mistakes to avoid

- Do not write ET and CET as the same thing - CET includes radiation effect.
- Do not measure solar incidence angle from the surface itself in the cosine law; use the angle from the surface normal.
- Do not claim air movement always cools by lowering air temperature - it improves convection/evaporation and changes perceived comfort.
- Do not use KHANA rules as universal modern codes - state the Bengal/Bangladesh climatic context.
- Do not draw shading devices without showing sun direction / blocked ray.
- In ventilation numericals, convert L/s to m³/s before using ACH formula.
- Always state units.

LECTURE 8

Source Basis & Credit

This note was built from the user-uploaded course materials. The wording has been organized for exam use, while preserving the class framing and correcting ambiguous points where necessary for technical accuracy.

Source group	Use in this note
Section A Lecture 1-11 slides	Primary structure, definitions, design points and examples.
Section A class notes (Shahriar Sir)	Cross-check of lecture emphasis and class terminology.
Student Question Bank Analysis	Lecture-wise matching question list and exam emphasis.
Regular Final Term Question Bank 2016-2024	Primary exam-question basis.
Manual of Tropical Housing and Building	Reference verification and conceptual background.
A Text Book of Climate & Design for the Built-form Designers	Reference verification, from which many course themes are derived.

Credit Made by 2223033